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E. Wolfe, R. W. Spekkens & The Postdoc Hiring Committee,
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To the Postdoc Hiring Committee at PI:

I am writing to express my interest in joining the research group of E. Wolfe and R. W. Spekkens as a post-doctoral student to work on the "Quantum Causal Inference Initiative". I am currently a PhD Candidate under the supervision of R. W. Spekkens in the Quantum Foundations group at PI and I expect to graduate April of 2023.

Causal Modelling and Quantum Foundations

For a variety of reasons, scientists are typically limited in their ability to interact with and collect empirical data about the complex, multipartite systems they care about. It is therefore necessary to develop tools which can make inferences about the structure of the whole system in the presence of partial information, and to furthermore characterize the intrinsic limitations of these tools.

Causal modelling, which is concerned with the structure of causal relationships between various subsystems, is a prominent example of such a tool. In recent years, causal modelling has has proven to be a valuable tool to the field of quantum foundations as it provides a framework for clarifying the fundamental differences between classical and quantum correlations (Wood and Spekkens 2015).

Past Research

I began my academic journey working with E. Wolfe on applying the Inflation technique (Wolfe, Spekkens, and Fritz 2019) to a particular causal structure involving a triple of pair-wise common causes called the triangle scenario (Fraser and Wolfe 2018). In that project, we derived inequalities necessarily satisfied by correlations achievable using classical common causes that are additionally violable by correlations achievable using quantum common causes, thus providing further evidence for the utility of causal modelling in quantum foundations. Afterwards, I developed a combinatorial approach to the problem of classical causal compatibility which is capable of generating a complete hierarchy of inner approximations for the space of compatible

correlations (Fraser 2020), complementing the outer approximations offered by the Inflation technique.

As a result of these experiences, it became clear to me that developing tools for quantum causal modelling would be a natural next step. Given that solving the classical marginal problem is a critical step for the Inflation technique, I became interested in tackling the analogous problem for quantum states known as the quantum marginal problem. In addition to its connection with quantum causal models, there are well-established connections between the quantum marginal problem and other topics such as quantum many-body physics, quantum error correction, quantum entanglement and quantum entropic inequalities.

Over the last two decades, tremendous progress has been made toward solving the quantum marginal problem for the special case of disjoint subsystems (Klyachko 2006). These advances were made possible by deep connections between representation theory and convex geometry (Walter et al. 2013). Unfortunately, the kinds of marginal problems encountered in causal modelling typically involve marginals pertaining to overlapping subsystems and thus new insights would be needed to make progress.

In my most recent paper (Fraser 2022), I managed to construct a hierarchy of necessary inequalities for the compatibility of an arbitrary collection of quantum marginals. Furthermore, using quantum state estimation techniques (Keyl 2006), I proved the sufficiency of this hierarchy. While there are numerous potential applications of these tools, I am most eager to apply them to problems related to quantum causal modelling.

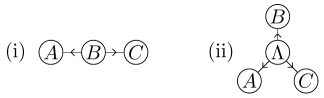
Future Research

One immediate application for the results of my recent paper to the topic of quantum causal modelling, at least under the assumption of known (but arbitrary) Hilbert space dimension for latent sources, is to construct an analogous convergent hierarchy of necessary inequalities for the compatibility of a tripartite quantum state with the quantum triangle scenario. Using existing results from the representation theories of the symmetric and unitary groups, it might be possible to analytically evaluate these constraints for large degree.

Additionally, while techniques developed for the quantum marginal problem are applicable to problems in quantum causal modelling, the reverse is also true: results from quantum causal modelling can be applied to the quantum marginal problem. To demonstrate this, consider that a tripartite quantum state, ρ_{ABC}, is compatible with the causal hypothesis that system B is the complete common cause of A and C, if and only if its quantum conditional mutual information vanishes (Allen et al. 2017). This condition is also equivalent to the condition that ρ_{ABC} can be decomposed by splitting system B such that $\rho_{ABC} \cong \bigoplus_i p_i(\rho_{AB_i^L} \otimes \rho_{B_i^R C})$ (Hayden et al.

2004). At the level of the quantum marginal problem, this condition implies that the tripartite state ρ_{ABC} can be uniquely reconstructed from its two-body marginals ρ_{AB} and ρ_{BC} (Zhang 2013), which in turn, constitutes a sufficient condition for the compatibility of ρ_{AB} and ρ_{BC}.

A direct consequence of this observation is a form of quantum advantage for causal inference (in a similar spirit to (Ried et al. 2015)), albeit in the restricted setting of causal inference starting from only marginal information (Budroni, Miklin, and Chaves 2016). Given any compatible pair of classical marginal distributions, p_{AB} and p_{BC}, it is impossible to distinguish between (i) the causal hypothesis that B is the complete common cause of A and C, and (ii) any saturating causal hypothesis, e.g., the hypothesis that a fourth system, Λ, is the complete common cause of A, B and C.



The reason for this indistinguishability is simply that there always exists a joint probability distribution, defined by $q_{ABC}(a,b,c) = p_{AB}(a,b)p_B^{-1}(b)p_{BC}(b,c)$, that is both compatible with hypothesis (i) and has marginal distributions p_{AB} and p_{BC}. This universal property of probability distributions is termed "adhesivity" in (Budroni, Miklin, and Chaves 2016).

In contrast, given a compatible pair of marginal quantum states, ρ_{AB} and ρ_{BC}, it is sometimes possible to distinguish causal hypothesis (i) from (ii). For instance, it can be shown that no tripartite quantum state compatible with causal hypothesis (i) has the bipartite marginals $\rho_{AB} = \rho_{BC} = \frac{1}{3}|00\rangle\langle 00| + \frac{2}{3}\left(\frac{|01\rangle+|10\rangle}{\sqrt{2}}\right)\left(\frac{\langle 01|+\langle 10|}{\sqrt{2}}\right)$. However, the W-state, $|W\rangle_{ABC} = \frac{1}{\sqrt{3}}(|001\rangle + |010\rangle + |100\rangle)$, which has the aforementioned bipartite marginals, is evidently compatible with causal hypothesis (ii).

More generally, it is worth investigating the distinguishability relations between causal models in the presence of only marginal information and to identify examples of discrepancies between classical and quantum causal models analogous to the above example.

In conclusion, there are a number of connections between the quantum marginal problem and quantum causal modelling worth investigating. Given my previous experiences working on both topics, I believe I am uniquely qualified to investigate these

connections further.

Sincerely,

Thomas C. Fraser

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